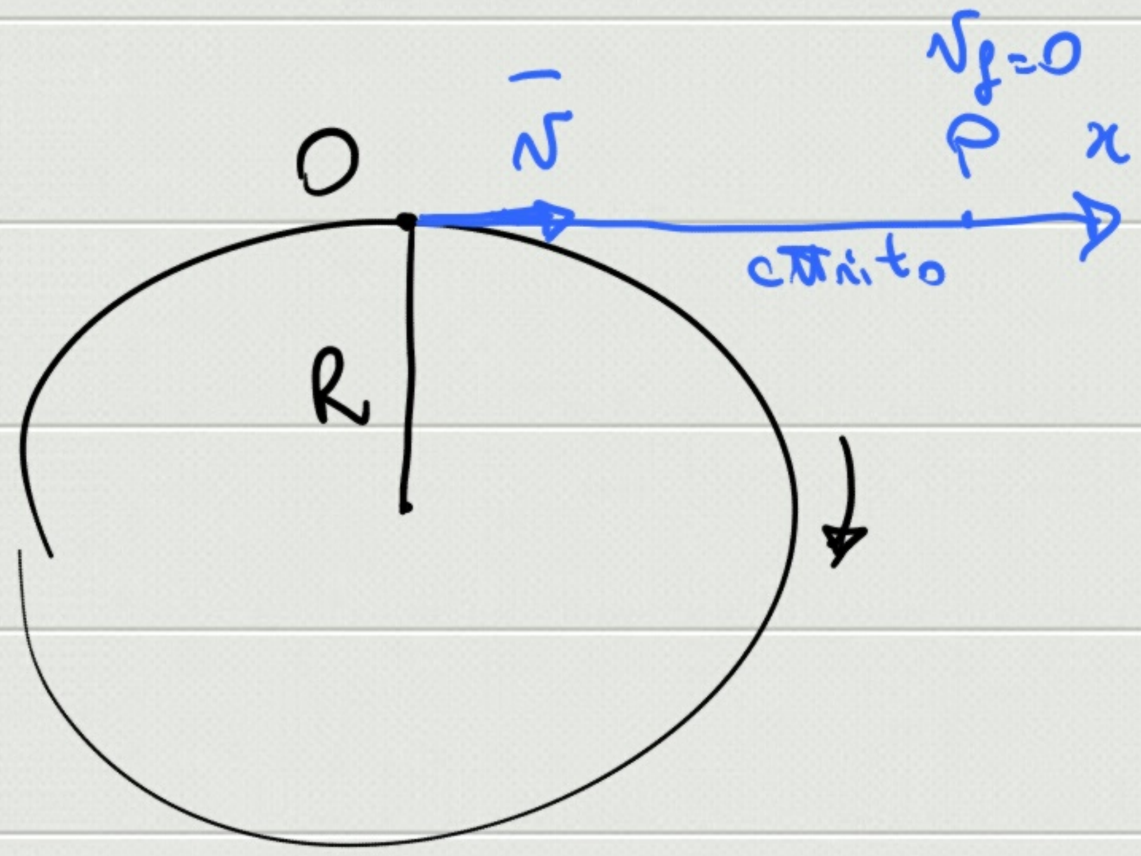


$m = 3.6\ kg$

$v_0 = 0$

$F_T = cost = 6.7\ N$

$R = 1.5\ m$

liscio

$v_f = 0 \quad OP = d = 8.4\ m$

$F_N\ (3\ giri) = ?$ $\quad \mu_d = ?$

- $v^2 = v_0^2 + 2 a_T \ell \quad \rightarrow \ell = 6\pi R \quad a_T = \frac{F_T}{m}$
- $\omega^2 = \omega_0^2 + 2\alpha \Delta\theta$
- $\begin{cases} \ell = \frac{1}{2} a_T t^2 \\ v = a_T t \end{cases}$
- $\begin{cases} \Delta\theta = \frac{1}{2}\alpha t^2 \\ \omega = \alpha t \end{cases}$

$$v^2 = 2\frac{F_T}{m} 6\pi R \implies v = \sqrt{12\pi \frac{F_T R}{m}}$$

$$F_N = m\frac{v^2}{R} = \frac{\cancel{m}}{\cancel{R}} \cdot 12\pi \cancel{R} \frac{F_T}{\cancel{m}} = \underline{\underline{12\pi F_T}} = 253\ N$$

$\Delta\theta = 6\pi\ rad \qquad \alpha = \frac{a_T}{R} = \frac{F_T}{mR}$

$$\Rightarrow t^2 = \frac{2\Delta\theta}{\alpha} = \frac{12\pi}{F_T} mR$$

$$\omega^2 = \alpha^2 t^2 = \frac{F_T^{\cancel{2}}}{m^2 R^2} \cdot \frac{12\pi}{\cancel{F_T}} \cancel{m} \cancel{R} = 12\pi \frac{F_T}{mR}$$

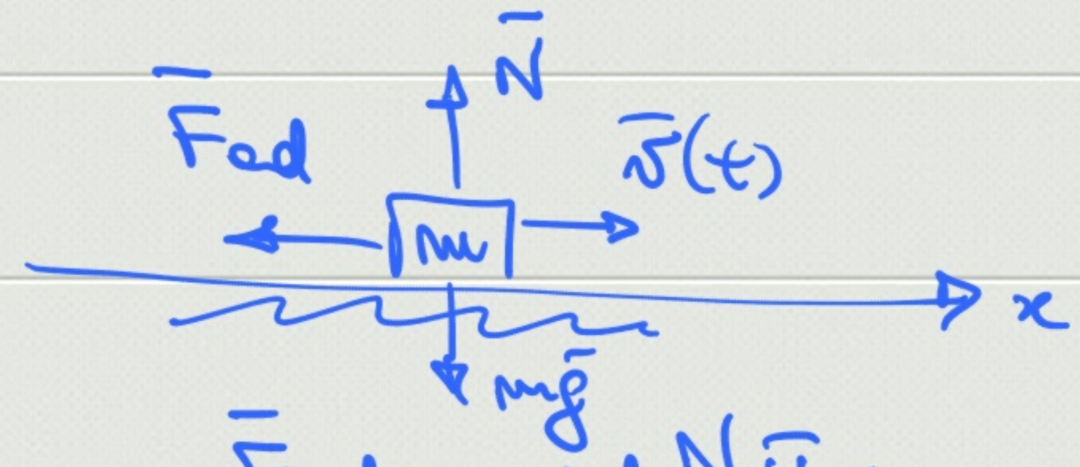
$$F_N = m\omega^2 R = \underline{\underline{12\pi F_T}}$$

$$- \quad 0 = v^2 - 2\mu g d \quad *$$

$$- \quad 0 = \cancel{v^2} - 2 a_T d$$

$$- \quad 0 = \cancel{v^2} + 2 a_T d$$

$$- \begin{cases} a = -\mu g \\ 0 = v - at \\ d = \frac{1}{2} a t^2 \end{cases} \quad *$$



$$\bar{F}_{ad} = -\mu_d N \bar{u}_v$$

$$\overset{!}{=} -\mu_d m g \, \bar{u}_v$$

$$\bar{F}_{ad} = m\bar{a}$$

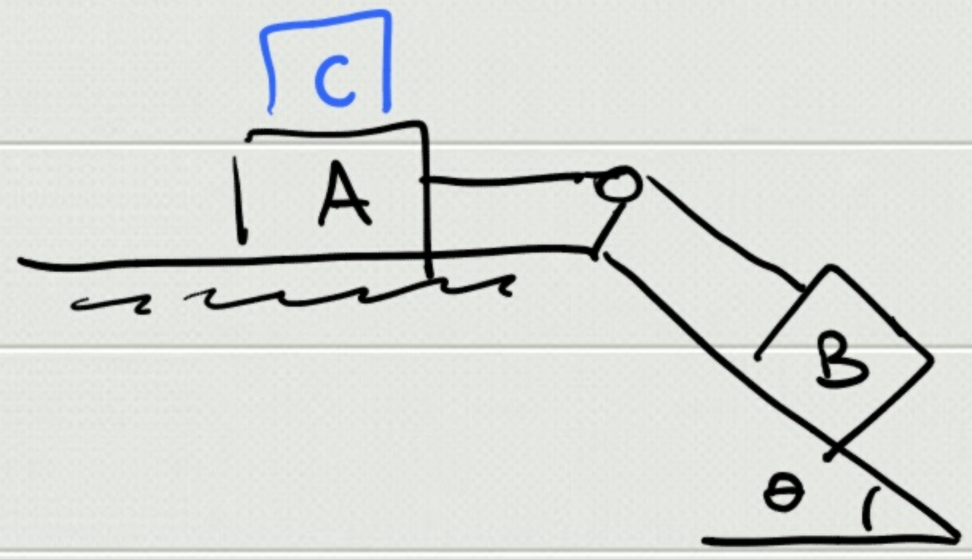
$$\Rightarrow \boxed{a = -\mu_d g}$$

$$v_f^2 = v_i^2 + 2a\Delta x \rightarrow d$$

$$v_f \rightarrow 0, \quad v_i \rightarrow v, \quad a \rightarrow -\mu g$$

$$0 = v^2 - 2\mu g d \Rightarrow \mu_d = \frac{v^2}{2gd} = \frac{6\pi F_T R}{mgd} =$$

$$= 0.64$$



$m_A = 10\ kg$

$m_B = 6\ kg$

$\theta = 30°$

$\mu_s = 0.25 \quad \mu_d = 0.2$

$m_{C,min} = ?$: eq. statico , $T = ?$

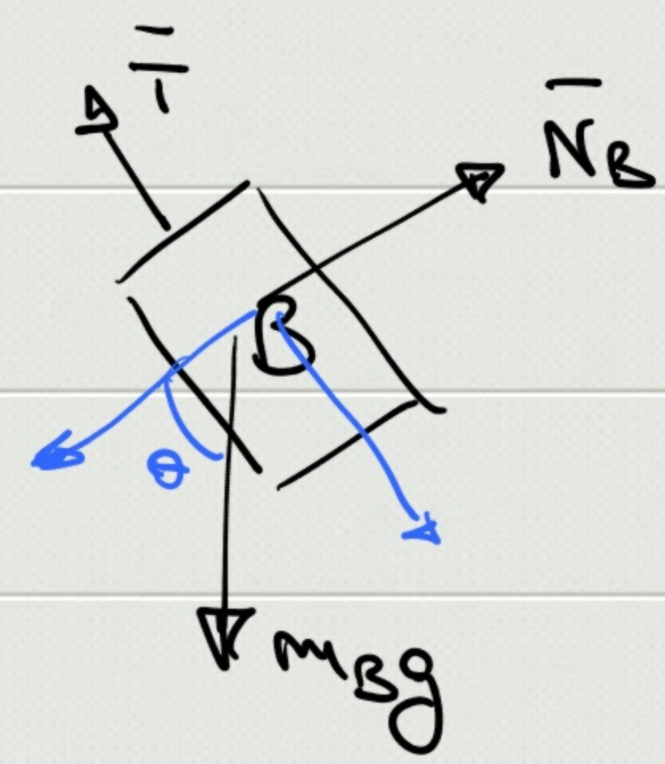
$T = m_B g \sin\theta$ *

$T - F_{as} = 0$ *

$T - F_{as} \leq \mu_s (m_A + m_C) g$

$F_{as} \leq \mu_s (m_A + m_C) g$ *

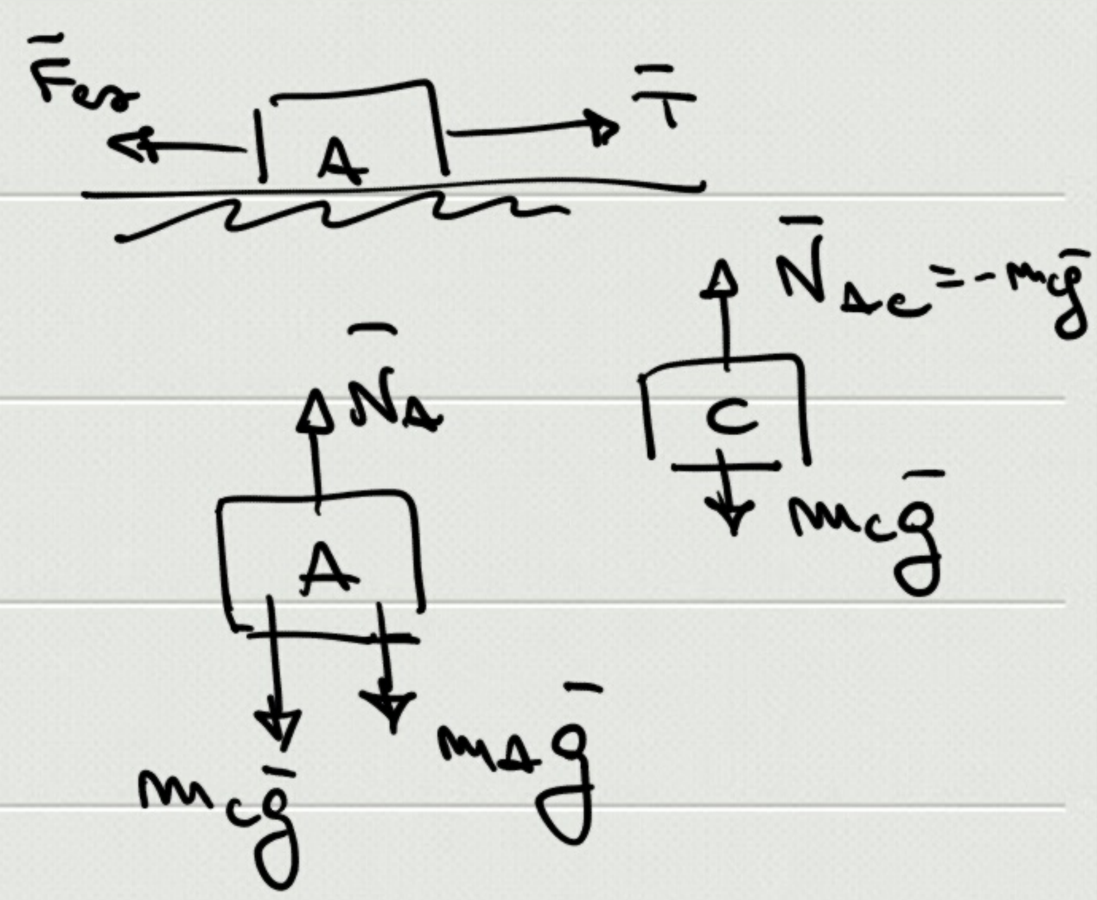
$F_{as} = \mu_s m_A g$



$T - m_B g \sin\theta = 0$

$\Rightarrow T = m_B g \sin\theta = 29.4 N$

$F_{as} - T = 0 \Rightarrow F_{as} = T$



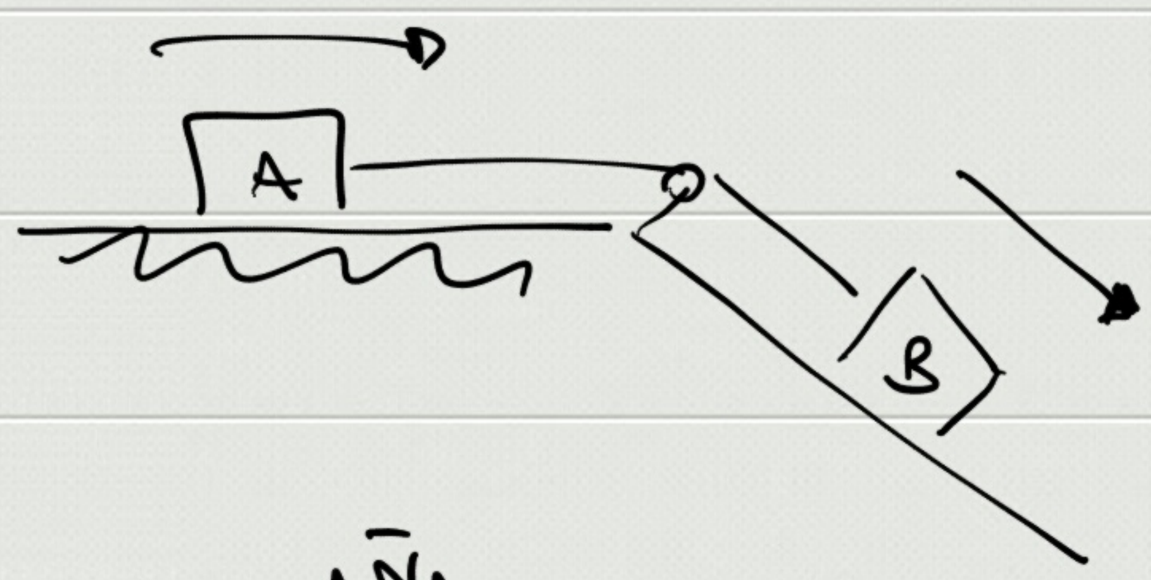
$F_{as} \leq F_{as,max} = \mu_s N_A$

$\Rightarrow N_A = (m_A + m_C) g$

$\Rightarrow F_{as} \leq \mu_s (m_A + m_C) g$

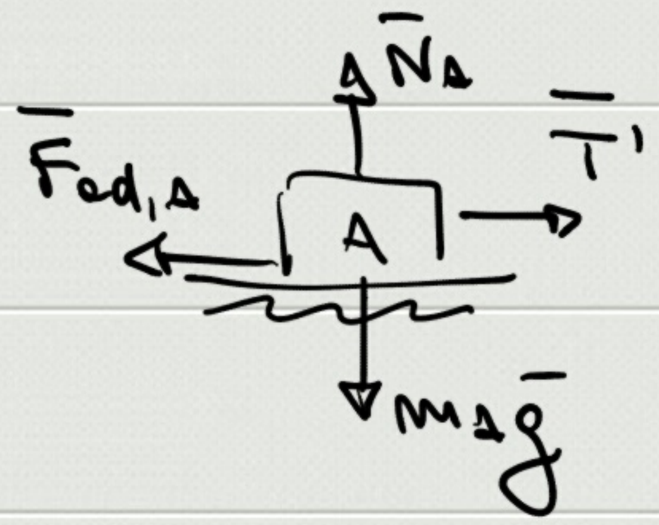
$$F_{as} = m_B g \sin\theta \le F_{as,max} = \mu_s (m_A + m_C) g$$

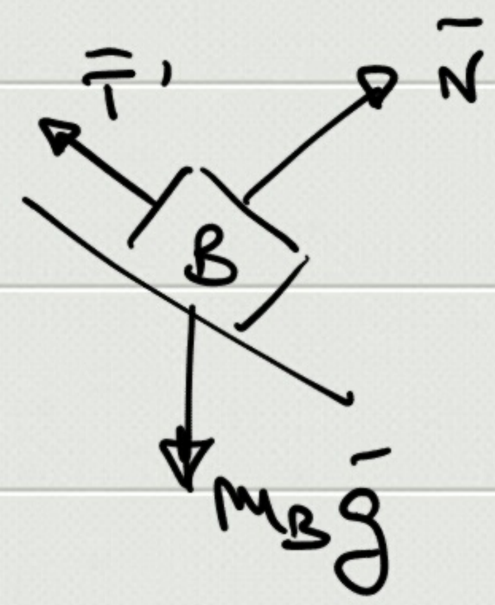
$$\Rightarrow m_C \ge \frac{m_B \sin\theta}{\mu_s} - m_A = m_{C,min} = 2\,kg$$



$a_A = ?$

$T' = ?$





$$\begin{cases} B: & m_B g \sin\theta - T' = m_B a_B \\ A: & T' - F_{ad,A} = m_A a_A \end{cases}$$

$a_A = a_B = a$

$$\begin{cases} m_B g \sin\theta - T' = m_B a \\ T' - \mu_d m_A g = m_A a \end{cases}$$

$$(m_B \sin\theta - \mu_d m_A) g = (m_A + m_B) a$$

$$a = \frac{m_B \sin\theta - \mu_d m_A}{m_A + m_B} g = 2.45 \ m/s^2$$

$$T' = m_A a + \mu_d m_A g =$$

$$= \frac{m_A m_B}{m_A + m_B} (\sin\theta + \mu_d) g = 14.7 \text{ N}$$